



Background

- In order to minimize waste and cost, clinical review of oral oncolytics often warrants sending partial fills due to scans, laboratory measures, or dosing regimen changes.
- It is important to consider the stability and appropriate storage of medications to ensure the patient is still receiving a viable product.
- Manufacturer labeling often has limited information, making it difficult to extrapolate the true necessity to keep medications in the original container.
- This project was designed to help oncology pharmacists better assess when it is appropriate to open bottles and/or packages to prevent waste and minimize cost.
- Thorough analysis of each medication via tertiary drug references, package labeling, and outreach to manufacturer will occur as applicable to identify best practice for each individual product.

Objectives

- To create a systematic workflow for the specialty pharmacy to utilize in determining whether a medication may be opened

Methodology

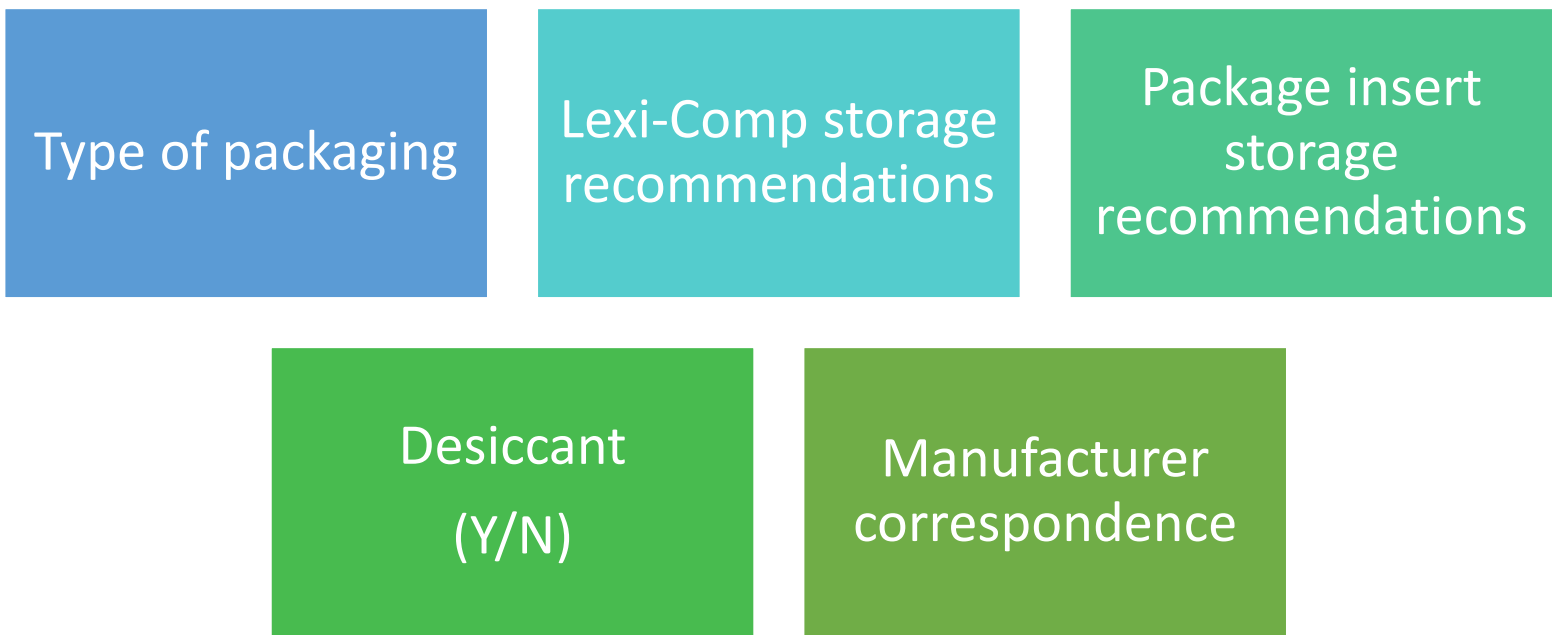
Identify Medications	• Initial phase was to conduct a retrospective review of all the medications that the pharmacy had historically opened
Conduct Research and Outreach	• Various students, residents, and pharmacists worked to compile data over time • Resources included: • Package insert • Medication guide or patient guide • Outreach to manufacturers • Requested written correspondence from manufacturers, if available, or alternatively documented phone calls • Assess whether medication contains a desiccant, any open dish studies, and any literature regarding use outside of the original container (with and/or without desiccant put into the alternative source)
Compile Findings	• All medications were put into a consolidated spreadsheet

Medication Use Evaluation (MUE)

- MUE identified over 160 medications to review
- 50% of applicable medications had been historically opened
- Equivalent to 36% of hematology/oncology medications

Initially reviewed and approved:	Second round reviewed and approved:	
Apalutamide (Erleada)	Niraparib and Abiraterone Acetate (Akeega)	
Axitinib (Inlyta)	Belzutifan (Welireg)	Nirogacestat (Ogsiveo)
Dabrafenib (Tafinlar)	Capivasertib (Truqap)	Asciminib (Scemblix)
Enzalutamide (Xtandi)	Fruquintinib (Fruzaqla)	Alectinib (Alecensa)
Pacritinib (Vonjo)	Ivosidenib (Tibsovo)	Adagrasib (Krazati)
Venetoclax (Venclexta)		

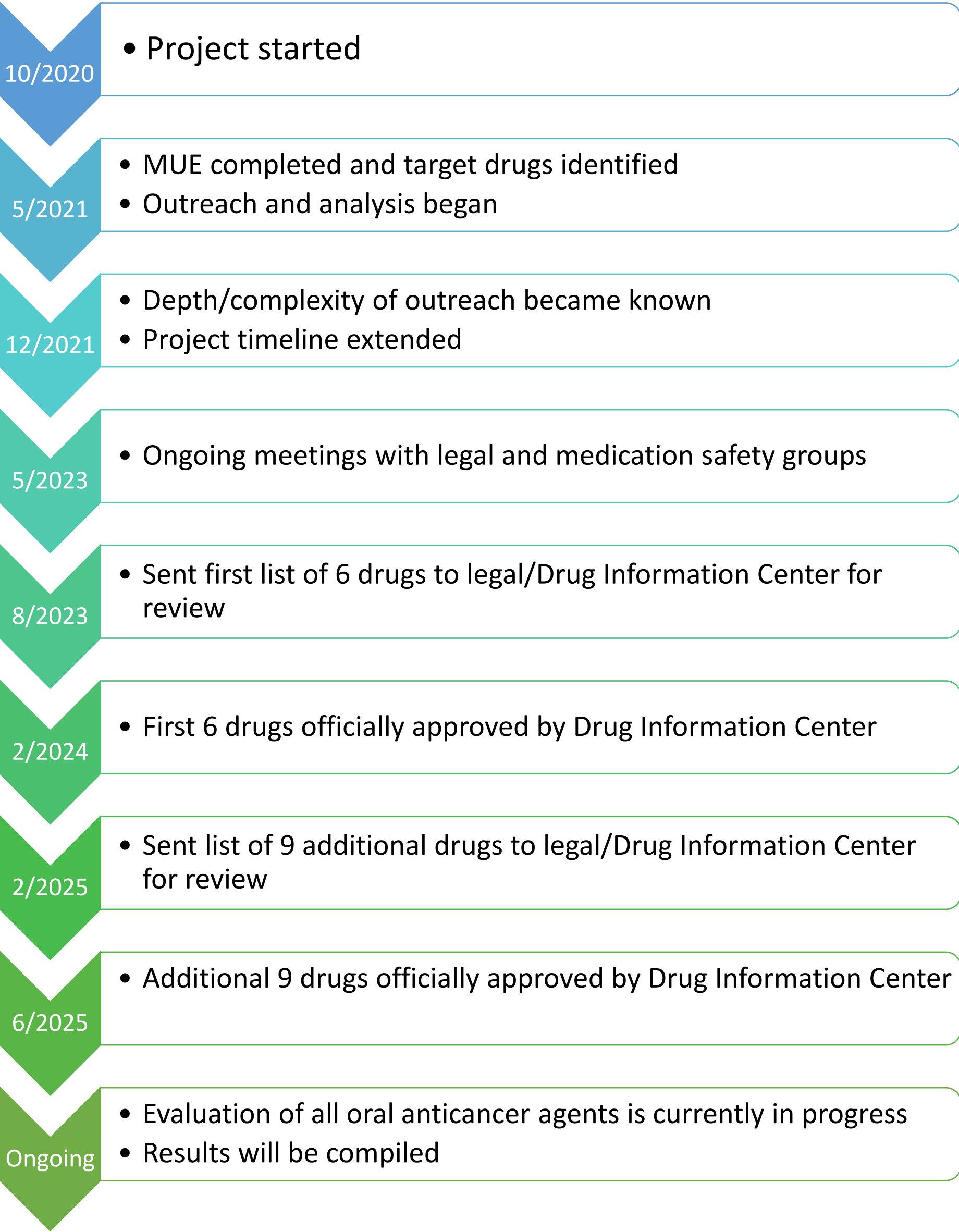
Data Collection



Prescribing Information and Excel Template

Storage and Handling						
Store at 20 °C to 25 °C (68 °F to 77 °F); excursions permitted to 15 °C to 30 °C (59 °F to 86 °F) [see USP Controlled Room Temperature].						
Store in original package to protect from light and moisture. Do not discard desiccant.						
Brand Name	Generic Name	Desiccant (Y/N)	Additional Comments	Package Size	Historically Opened Bottle (MUE) / uneven multiple of blister pack	Recommendations - OK TO OPEN?
Erleada	Apalutamide	Y	Only good for 10 days in pill boxes ; advise against this based on manf correspondence and studies conducted	#120	Y	Y - BUT NOT PILL BOXES

Timeline



References

1. Erleada® (apalutamide) [package insert]. Horsham, PA: Janssen Pharmaceutical Companies; 2019, revised 2024.

2. Akgöl K, van Merendonk LN, Barkman HJ, van Balen DE, van den Hoek HL, Klous MG, Hendriks JJ, Huitema AD, Beijnen JH, Nuijen B. Redispensing of expensive oral anticancer medicines: A practical application. J Oncol Pharm Pract. 2024 Apr;30(3):519-526. doi: 10.1177/10781552231176199. Epub 2023 May 16. PMID: 37192749.

3. Waterman KC, MacDonald BC. Package selection for moisture protection for solid, oral drug products. J Pharm Sci. 2010 Nov;99(11):4437-52. doi: 10.1002/jps.22161. PMID: 20845442.

4. Sato K, Inaoka N, Kodama Y, Muro T, Nakamura T, Sasaki H, Kitahara T. [Influence of Storage Conditions after One-dose Packaging on Stability of Magnesium Oxide Tablets]. Yakugaku Zasshi. 2018;138(11):1435-1441. Japanese. doi: 10.1248/yakushi.18-00024. PMID: 30381651.

5. Trovato JA, Tuttle LA. Oral chemotherapy handling and storage practices among Veterans Affairs oncology patients and caregivers. J Oncol Pharm Pract. 2014 Apr;20(2):88-92. doi: 10.1177/1078155213479417. Epub 2013 Mar 19. PMID: 23512268.

6. Zerillo JA, Goldenberg BA, Kotecha RR, Tewari AK, Jacobson JO, Krzyzanowska MK. Interventions to Improve Oral Chemotherapy Safety and Quality: A Systematic Review. JAMA Oncol. 2018;4(1):105-117. doi:10.1001/jamaoncol.2017.0625